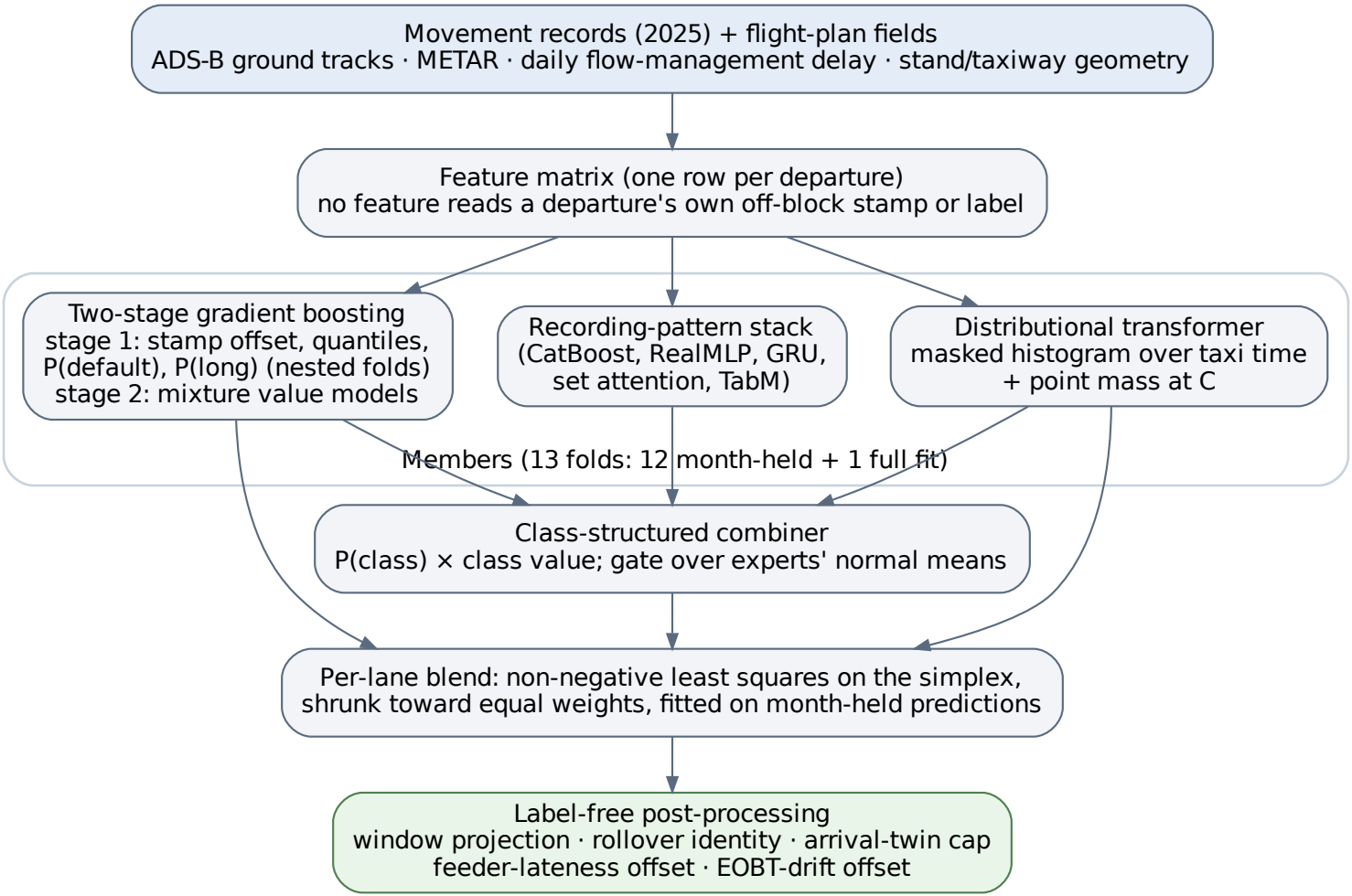
Movement records (2025) + flight-plan fields
ADS-B ground tracks · METAR · daily flow-management delay · stand/taxiway geometry
Feature matrix (one row per departure)
no feature reads a departure's own off-block stamp or label
Two-stage gradient boosting
stage 1: stamp offset, quantiles, P(default), P(long) (nested folds)
stage 2: mixture value models
Recording-pattern stack
(CatBoost, RealMLP, GRU, set attention, TabM)
Distributional transformer
masked histogram over taxi time
+ point mass at C
Members (13 folds: 12 month-held + 1 full fit)
Class-structured combiner
P(class) × class value; gate over experts' normal means
Per-lane blend: non-negative least squares on the simplex,
shrunk toward equal weights, fitted on month-held predictions
Label-free post-processing
window projection · rollover identity · arrival-twin cap
feeder-lateness offset · EOBT-drift offset